CHAPTER 5
HARD DISK DRIVES

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Chapter Objectives

Upon completion of this chapter, the student will be able to:

- Explain how data is physically & logically organised on a disk.
- Distinguish between different hard disk technologies.
- Manage disks using Operating Systems (Windows & MS DOS).
- Install IDE/ATA & SATA hard disk drives.
- Install OS and application SW.

Sub-topics

5.0 Introduction
5.1 Storage of data on hard disk drive
5.2 Hard disk drive technologies
5.3 MS DOS commands for Disk management
5.4 Formatting disks
5.5 Optimising HDDs

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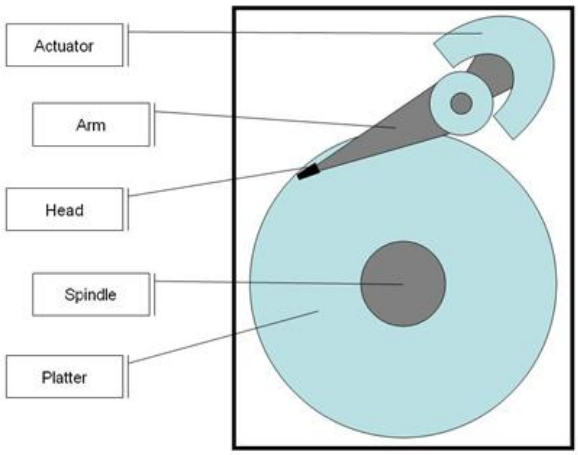
5.0 Introduction

HDD (hard disk drive) or simply **hard disk** is the main secondary storage device with the biggest capacity. It is a permanent, magnetic disk which is either internal (in this Chapter) in bays or external (in later Chapter) in housing. It is used to store OS, applications software and user data files (e.g. word documents, spreadsheets, databases, music).
The first HDD had storage capacity of only 5 MB, but today the capacity has increased to hundreds of GB or even TB (terabytes). HDD uses random access (compared to sequential access for magnetic tapes).

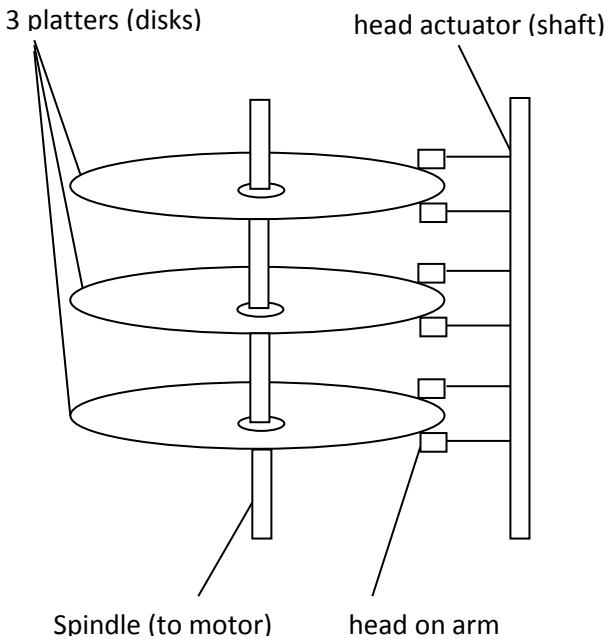
External parts of HDD

-Label on top of HDD has information which include:
HDD technology, manufacturer of HDD and jumper settings for installation.
-Major manufacturers of HDDs are:
Western Digital, Seagate, Samsung, Toshiba, Fujitsa & Hitachi.
-Jumpers are wires which close electrical circuits, allowing the electricity to flow through specific sections of the PCB.
See installation of HDD later.
-Interfaces of HDD to PC (used for installation) are: power cable connector, data connector (IDE connector) & jumper pins.

Internal parts of HDD



HDD: Top view



HDD: Side view

Platters & Spindle

- **Platters (disks)** have 2 polished smooth surfaces (top & bottom) made of high precision aluminium alloy or glass, and each surface is coated with a magnetic material.
- They are connected to a **spindle** (clamping mechanism) and spin (rotate) at high speeds using a high speed linear motor.
- Most hard disks have **multiple platters** in order to increase the amount of information to be stored.

Heads & Head actuator

- There is one read/write **head** per surface of disk i.e. each platter has 2 heads (Top & Bottom) e.g. 3 platters with 6 heads.
- **Head actuator (shaft)** controls the movement of the heads (positions the heads with precision).
- There is one **arm** per head, and all of these (heads, arms & actuator) are lined up to form one system.

How HDD works

-The actuator moves the arm to the spindle in order to position the head over the platter where data is written or read.
-The heads fitted on arms are able to move in and out over the disks, but not touching their surfaces.
-When writing data, the head produces a magnetic field that magnetizes the surface of the platter to store data.
-The magnetic fields produced by platter can also be read by the head to retrieve data.

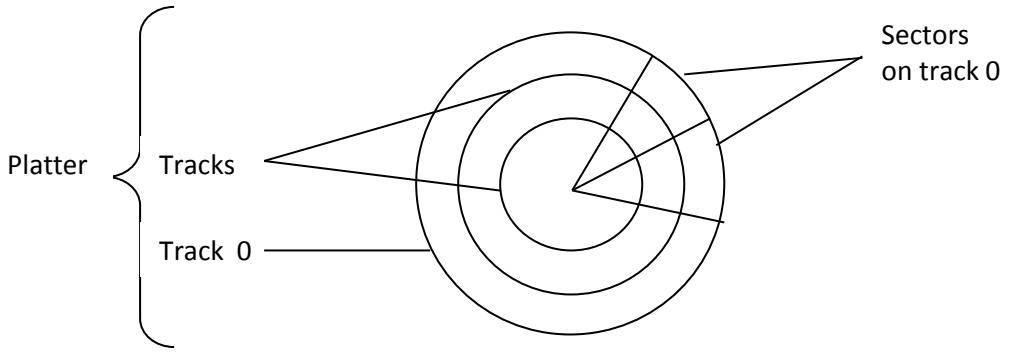
Note:
- **HDD controller** with logic (electronic components) to enable CPU communicate with HDD.
- The maximum number of heads for all HDDs should be 16, though some HDDs have only one platter with up to 12 heads.
- Some expensive HDDs (for very large computer systems) have one read/write head per track.
- Some HDDs use **voice coil** technology (similar to loud speakers) with extra one or two heads for accuracy of the arm position (which may have odd number of heads).
- Some HDDs use **sector translation** technology that allows them to have more than two heads per platter.

Compare FDD (Floppy Disk Drive)
- FDD also known as **floppy disk** or simply **diskette;** was another secondary storage device used in older PCs, but has now become obsolete.
- FDD is also permanent and magnetic disk placed internally in bays and of 2 main form factors (3½ inch & 5¼ inch).
- FDD used to be standard media for transferring and exchanging data or information between PCs, but now taken over by **flash disks** e.g. device drivers not in Windows and updates (e.g. for BIOS and Windows).
- FDD bootable disk (start-up disk) when OS on HDD was corrupted and cannot boot.
- Also has: magnetic heads (1 or 2), disk, spindle (clamping mechanism), motor (rotates disk), actuator (positions heads), sensor (detects rotational position) and FDD controller (generates signals for heads).

Problems of FDDs
1. Radial misalignment.
2. Head rotates slightly over its axis.
3. Magnetic sensor out of position.
4. Allergic to computer virus attack.

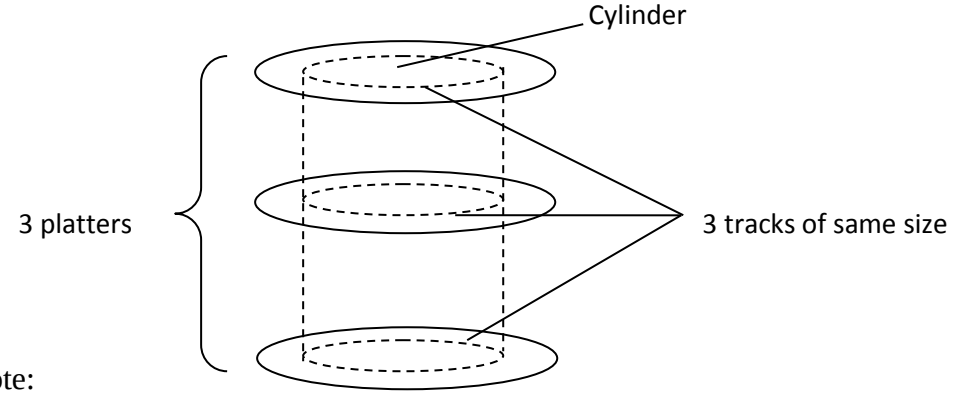
5.1 Storage of data on hard disk drive
Consider:
1. Physically 2. Logically

Physically
Physical storage refers to HDD structure i.e. platters (disks) are demarcated into **tracks** and **sectors**.



Each of the:
- **Platters (disks)** consists of 2 disk surfaces (top and bottom).

- **Surfaces** consists of hundreds of **tracks,** with outermost as **track 0 (zero)**.
- **Tracks** (concentric circles) are circular and consist of **sectors**.
- **Sectors** (arcs) is a smallest unit of storage space on disk of fixed capacity 512 bytes.
- **Blocks (clusters)** is a group of sectors (1 or more) e.g. MS DOS block of data is 1 to 8 sectors.
- **Cylinders** consists of a set of tracks of the same diameter on each platter.



Note:
- To refer to a sector, specify: 1. disk surface 2. Track number 3. Sector itself.
- **Boot strap loader** in ROM BIOS is copied during formatting to **sector 0 (zero)**.
- BIOS limitations are 63 sectors per track and 1024 cylinders per HDD.
- Magnetic density increases from outside sectors to the centre of platter i.e. near circumference (low density), but near centre (high density).
- Since sectors are not of equal physical length, **write pre-compensation** (using cylinders) is used so that all sectors are fixed at 512 bytes.
- **T**he capacity of HDD is determined by **CHS (Cylinders, Heads & Sectors per track) values** and system BIOS.
- CHS values limit the maximum size of a hard disk drive to 504 MB under the older PC operating systems.
- HDD capacity can be increased in 2 ways:
  1. Bypass system BIOS and instead use BIOS of HDD.
  2. Change the way system BIOS routines are read.

Example
HDD has 1024 cylinders, 16 heads and 63 sectors per track. What is its capacity in MB?

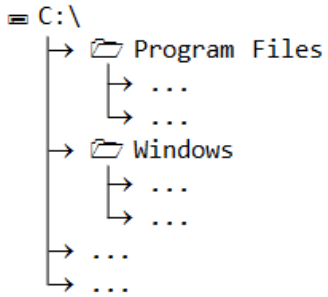
Solution
Remember: 512 bytes per sector and **tracks** = cylinders x heads
Capacity = cylinders x heads x sectors x capacity of sectors
Capacity = 1024 × 16 × 63 × 512 = 528,482,304 bytes = 504 MB

Logically
Logical storage refers to file system on the HDD i.e. how directories (folders), sub directories (sub folders) are used to organise files on a disk. Directories and Folders are the same, depending on OS used e.g. MS Dos, Windows or Linux. **File** is a group of related information called **records** on a secondary storage device (e.g. HDD). Compare directory (folder) and sub directory/sub folder, with filing cabinet and drawers of the cabinet.
Note:
- File names has a base name, a **file extension** and a unique **path name**.

- The **file extension** specifies the type of file e.g. **.txt** (text file), **.doc** (Word document), **.htm** (hypertext mark-up) or **.html** (hypertext mark-up language).
- The **path name** shows the path followed from root directory through subdirectories to the file. For example, **C:\users\computer\hardware.doc** refers to path name of file **hardware.doc** i.e. found in subfolder **computer**, subfolder **users** and drive C.

MS DOS
-Uses directories and subdirectories to organise files on disks.
-Each **directory** can contain other directories or subdirectories or files.
-Each **subdirectory** can contain other subdirectories or files.
-Also uses a **tree structure** of directories (subdirectories) to organise files.
-**Root directory** is the starting point (top-most directory) in the tree structure and contains all other directories (can also contain files).
-**Current directory** refers to the directory you are currently one logged on, and can be changed using a DOS command **CD**.
-Allows only 8 characters for a base name and 3 characters for a file extension.
-Recognises only a few extensions e.g. binary executable files **.com**, **.exe**, **.bat**.

DOS tree structure



See MS DOS commands to manage disks later.

Windows
-Uses folders and sub folders to organise files on disks.
-Each **folder** can contain other folders or sub folders or files.
-Each **sub folder** can contain other sub folders or files.
-Example of a folder, is C drive in **My Computer** represented as **C:** (letter & a colon).
-**Root folder** is the starting point (top-most folder) in the hierarchy and contains all other folders (can also contain files).
-**Current drive** refers to the drive you are currently logged on, and can be changed.
.**Windows file systems**
The 2 main types are:
1. FAT (File Allocation Table) 2. NTFS (New Technology File System)

The 2 types of FAT (File Allocation Table) are:
1. FAT16 (16-bit FAT) 2. FAT32 (32-bit FAT)

Note:
- The basic storage unit for drives is a sector, which can store between 1 byte and 512 bytes.
- It means any file less than 512 bytes is stored in one sector (1 file per sector).
- Number of sectors occupied depends on size of a disk e.g. diskette of 1.44 MB had 9 sectors.

FAT16 (16-bit FAT)
-Introduced in DOS 3.0; it is a simple file system used by small disks (e.g. former diskettes & flash disks) and simple folders.
-FAT16 came after FAT 8 and FAT12 File systems which became obsolete.
-FAT16 was used with older Windows systems i.e. from 3.x to 95, but also included in newer Windows OS to provide backward compatibility for upgrades.
-The 16-bit FAT address gives 2^{16} = 64,000 sectors @ 512 bytes.
This means, maximum hard drive partition = 64,000×512 bytes i.e. about 32 MB.
-Large HDDs have more than 32 MB because of they use 2 methods:
1. fdisk to get multiple partitions @ less than 32 MB.
2. Clustering i.e. combine a set of contiguous sectors into a single unit.

-For Windows NT (e.g. Windows 7), maximum size of FAT16 volume = 4,095 MB.

FAT32 (32-bit FAT)
Introduced in Windows 95 SP2 (Service Pack 2); and then used in Windows 98, 2000, Mellinium. Not used in newer Windows systems i.e. from Windows XP to Windows 7.
-It is more efficient in space allocation on large hard disk (512MB to 2TB).Why?
FAT32 uses smaller clusters than FAT16. All users can have access to all files on the hard disk drives using FAT32. Compare with NTFS later.

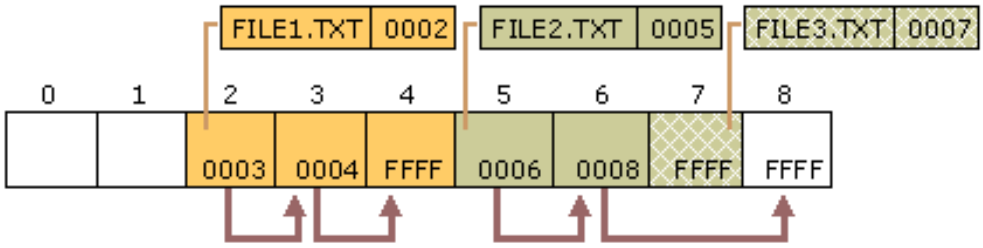
Structure of FAT
1. Boot sector i.e. 1st sector = 512 bytes
2. FATs i.e. FAT1 and FAT2 (duplicate)
3. Root directory

Boot Sector	FAT 1	FAT 2 (Duplicate)	Root Folder	Other Folders and All Files

Boot sector
Contains machine code to be loaded into RAM by BIOS (firmware) during booting.
It allows the boot process to load a program (e.g. OS) stored on the same storage device.
FATs
Each partition has two identical FATs (FAT1 and FAT2) stored near the beginning of the partition. Why? As a precaution, in case of disk damage (corruption) they can be compared.

Root directory
The 1st top-most directory in the tree structure and contains all other directories.

Files in FAT



The diagram above shows FAT files stored in 9 clusters (blocks).
Note:

- The folder entry points to the first cluster of the file.
- Each cluster contains a pointer (arrows) to next cluster or EOF.
- The end of the file cluster is indicated as EOF (End-Of-File).
- File1.txt is large with 3 clusters, File2.txt is a fragmented with 3 clusters (only 2 occupied) and File3.txt fits exactly in 1 cluster.

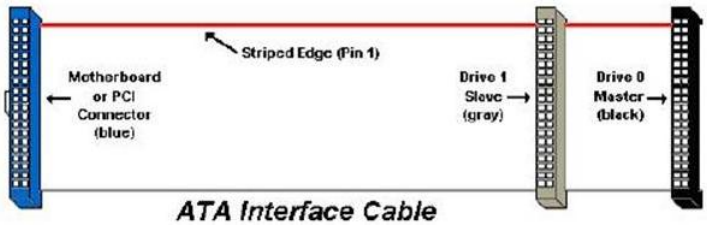
NTFS (New Technology File System)
Supports limited accounts i.e. not all users can have access to all files on the hard disk drives. For minimum capacity of 10MB and used in newer Windows systems i.e. from Windows XP to Windows 7.

5.2 Hard disk drive technologies
-The HDD technologies differ according to interfaces:
1. Data cables and their connectors
2. Power cable and their connectors
3. Motherboard connectors for data cables
4. Jumpers and their pins.

-The HDD technologies include:
1. IDE(Integrated Disk Electronics)
2. SATA (Serial ATA)
3. SCSI (Small Computer Systems Interface)

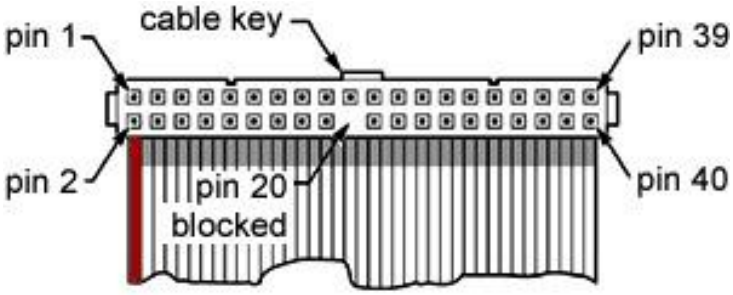
-Other improved HDD technologies are:
1. EIDE (Enhanced IDE)
2. PATA (Parallel ATA)
3. ATAPI (ATA for Packet Interface)

IDE (Integrated Disk Electronics)
-IDE also called **ATA (Advanced Technology Attachment)** by Western Digital (In Malaysia), used to be standard technology for desktop PCs.
-The parallel design of original IDE hard drives provided maximum speed of 133 mbps for ATA7.
-IDE have different nos. of heads, cylinders and sectors for the same IDE interface card.
-IDE is also referred to as ATA because the 2 are similar in the following ways:
1. Support up to 2 devices (master & slave), but of same type.
2. Data interface have 40 or 80 pins.
3. Bus width = 16 bits.
4. Controller chip on HDD.



IDE hard disk connectors IDE cable connectors

With label facing up, interfaces facing you and left to right are:
IDE cable connector (40 or 80 pins), jumper pins (6 to 10 pins) & power cable connector (4 pins). There are 3 connectors i.e. one at each end of the cable and the 3rd one about 2/3 from the motherboard connector. Pins on IDE cable connector are numbered from top to bottom and right to left i.e. pin 1 is top and on the side of power connector.



Advantages of IDE/ATA

1. Simple to install (set up).
2. Can handle large data.
3. Fast enough to boot.
4. Supports 2 HDDs.
5. Least expensive.

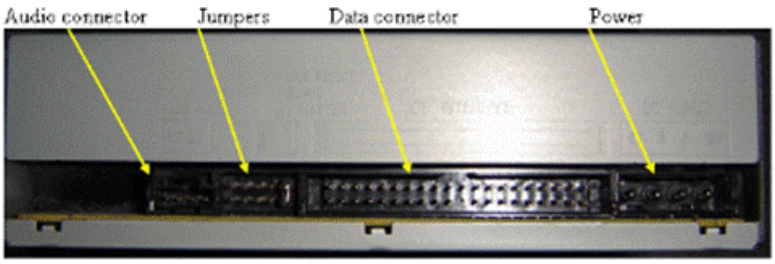
Note:

- EIDE also called ATA-2; extended the IDE design to allow two cables (primary IDE and secondary IDE), each with 2 drives.
- EIDE was an improvement on IDE; now replaced by better technologies e.g. SATA & SCSI.
- EIDE also had ATAPI (ATA Packet Interface) for CD drives (previously used custom-built interface card).

Advantages of EIDE

1. Increased data rates (throughput) i.e. 3 to 4 times faster than IDE.
2. Support of storage devices other than hard disk drives.
   Allowed mixture of disks (other types other than HDDs) i.e. CD-ROM, DVD-ROM & magnetic tape.
3. Supported up to 4 IDE devices instead of just 2 i.e. BIOS supports 2 controllers, each with 2 drives.
4. Supported backward compatibility i.e. EIDE catered for older IDE.
5. Support for hard disk drives larger than 528 MB.

Compare CD-ROM & DVD-ROM drive connectors



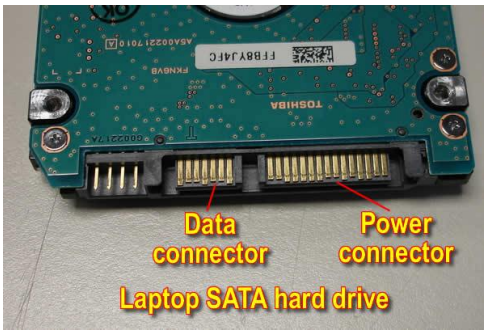
Disadvantages of EIDE

1. Cables are large.
2. Not fast as SATA & SCSI.

EDIE connector is built into the motherboard and requires no disk controllers in slots.
See installation of IDE/ATAdisks later.

SATA (Serial ATA)

It is an improvement of the older PATA (Parallel ATA), and communicates via a high-speed serial cable. SATA hard drives are quickly becoming the standard for both desktop and laptop PCs, because most recent motherboards support them.



SATA drive connectors SATA data cable SATA power cable

-With label facing down, interfaces facing you and left to right:
-Jumper block (factory use only), SATA data connector
(7 conductors) & SATA power connector (15 pins).
-SATA cable has 7 conductors and connects a single HDD to a single SATA connector on the motherboard i.e. SATA is **serial**, but IDE/ATA is **parallel**.
-Both ends of the cable (thinner and maximum of 1 m long) are identical so the cable may be connected either way round.
-The power cable provides 3.3 V, 5 V and 12 V power. Some use traditional power connectors, while others have both using adaptors.

Note:

- SATA uses much lower signal voltage levels, which allow faster signalling.
- It also reduces the electrical problems associated with IDE parallel transmission by using balanced, full-duplex serial transmission.
- SATA ideal data rates are 150 (version 1) to 300 mbps (version 2), compared to PATA maximum data rate of 133 mbps.

Advantages of SATA

1. Thinner longer serial cables.
2. Longer drive life.
3. Faster data rates (boots very fast).

Disadvantages of SATA

1. More expensive than IDE/ATA.
2. Need adapter card to convert IDE to SATA.

Larger IDE & Smaller SATA connector on MB

IDE converted to SATA using adapter card

What are the differences between IDE and SATA in terms of cables, speed, jumper settings and controllers?

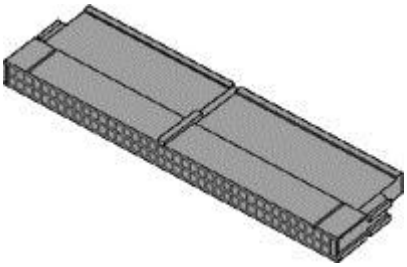
SCSI (Small Computer Systems Interface)
It was designed as a general purpose interface for a range of devices, but it is most commonly used for high-speed hard drives. Commonly used for network servers and high end powerful workstations (computers) e.g. Apple adopted SCSI as its expansion bus standard. This makes them to be more expensive than the other HDD technologies.

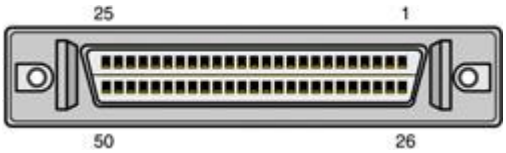
Older versions of SCSI
1. SCSI-1 (Standard SCSI) Original SCSI with 8-bit data transmission at 5mbps.
2. SCSI-2 (Fast SCSI) in 1989 with faster 8-bit data transmission at 10MBps.
3. SCSI-3 (Wide SCSI) Supported up to 15 devices and of speeds up to 2 mbps.
4. SCSI-3 (Ultra SCSI) in 1992 with improved speeds to 20MBps.

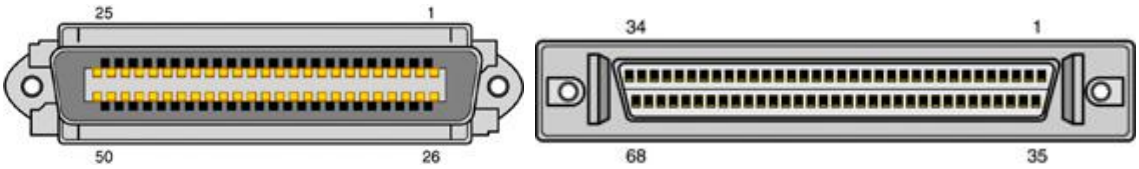
Newer versions of SCSI
1. Ultra-2 SCSI in 1997 which doubled speeds to 40MBps and 80MBps.
2. Ultra-3 SCSI in 1999 which again doubled speeds to 160MBps.
3. Ultra-320 SCSI in almost all current SCSI drives with speed of 320MBps.

Note:
- SCSI-1 supported 7, while SCSI-2 supported 15 peripheral devices using **daisy chain** connection.
- Some adapters allow multiple chains for even more devices i.e. connecting hard disks, printers and scanners.

SCSI cables and connectors
1) 50-pin (A-cables)
2) 68-pin (B-cable, P-cable & Q-cable)







See installation of SCSI devices later.

Factors to consider when shopping for a hard disk drive
1. Interfaces e.g. motherboard connectors, power supply connector
2. Hard disk capacity in GB
3. Data rate in mbps i.e. transfer speed and seek time.
4. Internal or external hard disk drive
5. Desktop or laptop hard disk drive

Internal hard disk drive is generally less expensive and does not require additional cables or separate power.

Hard drive specifications
1. Capacity
   Measured in GB (gigabytes) or TB (terabytes).
2. Speed of disk spins
   Measured in RPM (revolutions per minute).
   Has effect on how quickly data can be stored and retrieved.
3. Seek time (access time)
   Measure of how long it takes head to find data or location of free space.
4. Data transfer rate measured in mbps.
5. External interfaces such as eSATA, FireWire, SCSI or USB

5.3 MS DOS commands for Disk management

Classification of DOS commands
By location as:
1. Internal commands
   Stored in COMMAND.COM file and loaded into memory when PC boots.
   e.g. CLS, DATE, TIME, VER, EXIT,CD, DIR, COPY, DEL, MD, REN
2. External commands
   Exist as separate executable files on disk.
   e.g. CHKDSK, FORMAT, EDIT, LABEL, BACKUP, DISKCOPY

By action on:
1. Drives/directories e.g. DIR, CD, MD, RD, TREE
2. Disks e.g. FORMAT, CHKDSK, BACKUP, RESTORE, DISKCOPY
3. Files e.g. COPY, DEL or ERASE, REN, XCOPY, TYPE

Some common DOS commands
CD
Short form of CHDIR, used to display current drive letter and folder.
Also used to change the drive or directory that you want to work in.
CHKDSK
Refers to check disk, used to identify and correct certain hard drive errors.

Used to check a disk and display a status report with properties of disk eg serial number, volume label, memory and other properties along with errors if any.
CLS
Clears the screen of all previously entered commands and other text.
COPY
Copies one or more files from one location to another.
DATE
Used to show or change the current date.
DEL
Used to delete one or more files. It is the same as the ERASE command.
DIR
Used to display a list of files and folders contained inside in the current folder.
Also displays other important information e.g. hard drive's serial number, total number of files listed, their combined size, total amount of free space left on the drive etc.
DISPART
Used to partition disks. (See later).
EXIT
The exit command is used to end the command.com current session.
Short form of MKDIR command, used to create directory (make directory).
FORMAT
Used to format a disk. (See later).
LABEL
Used to show volume label of a disk, if it exists.
MEM
Displays different types of memory ie used and free conventional, upper, reserved and extended (XMS) memory.
MOVE
Used to move one or more files from one folder to another and also used to rename directories.
RD
Short form of RMDIR (Remove Directory) command. Used to delete a directory.
RENAME
Used to change the name of the individual file that you specify.
REPLACE
Used to replace one or more files with one or more other files.
RESTORE
Used to restore files that were backed up using the backup command.
TIME
Used to show or change the current time.
TREE
Used to graphically display the folder structure of a specified drive or path.
TYPE
Used to display the information contained in a text file.
UNDELETE
Used to undo a deletion performed with the MS-DOS delete command.
VER
Used to display the current MS-DOS version number.
VOL
Shows the volume label and serial number of a specified disk, assuming this information exists

Note:
- MS-DOS command line is not case sensitive e.g. **dir.** is the same as **DIR**.
  Compare with Linux which is case sensitive.
- The names of files or directories with more than two words should be put in quotation marks e.g. computer hardware as “computer hardware ".
- When a file or directory is deleted in the command line, it is not moved into the Recycle bin.

Syntax of DOS commands
Consists of:
1. Command name e.g. DIR, CD
2. Parameters e.g. colon: / (forward slash), \ (backslash),.. (two dots)
3. Switches e.g. /p (pause), /w (width wise), /v (volume)
4. Wildcards e.g. * (all characters),? (single character)

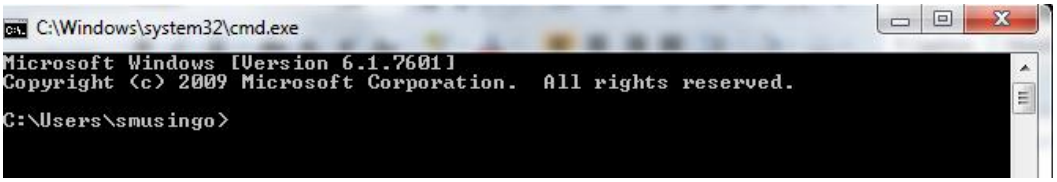
Examples
DIR command
DIR /p (forward slash) Pauses after displaying each screen.
DIR /w (forward slash) Displays in a wide format.
DIR/w/p (forward slashes) Displays in wide format and pauses.
DIR /? (Forward slash?) Find help (options) on command DIR.
DIR *.* (star dot star) Displays all files in current directory.

CD command
CD.. (2 dots) Moves one directory lower.
CD\ (backslash) Moves to root directory (usually drive C).
CD E: (colon) Change to directory E.

DEL command
DEL C: *.ext (star dot ext) Delete all files in drive C with extension .ext
DEL C: *.* (star dot star) Delete all files in drive C.
MS DOS prompts you to confirm the action. Delete? (Y/N)? Why prompt?

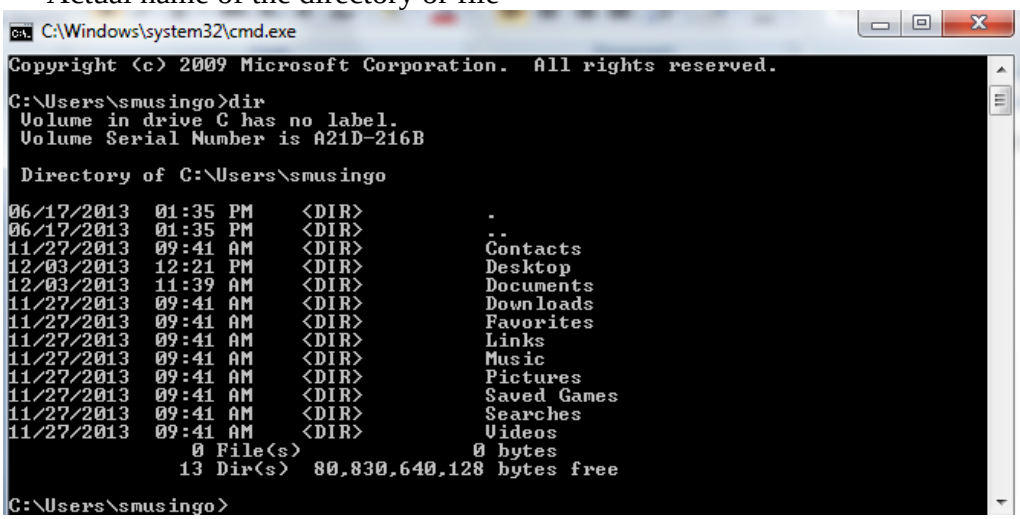
CHKDSK command
CHKDSK /f /r
Switches /f and /r attempt to fix file system errors.
When scan completes, bad sectors are marked and system automatically reboots.

Task 1: To enter Windows 7 Command Prompt
Click Start button and type **cmd** in search button.
OR
-Use Run utility ie Windows + R.
-Type **cmd** command in Run dialog box and press Enter.
-By default Windows opens user’s directory as seen below.
-The prompt changes to eg **C:\Users\smusingo>** (user directory).

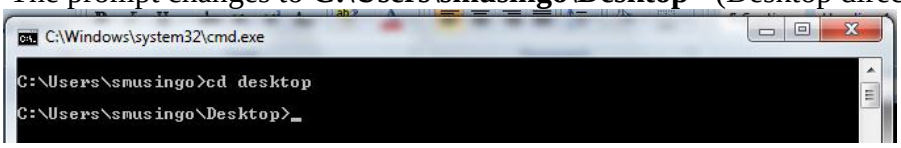


Task 2: To use DIR command
-Type **DIR** command at command prompt e.g. **C:\Users\smusingo>** (user directory).
-Useful information displayed include:

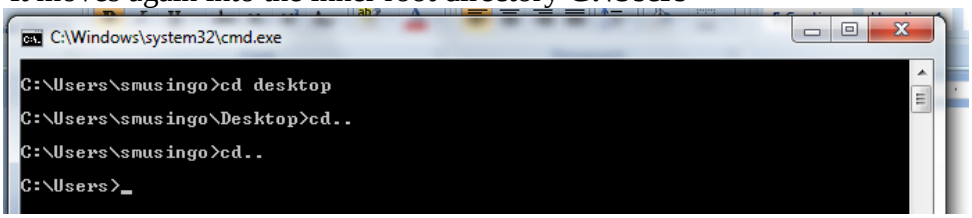
- Modification or creation date and time
- Indication if the listed item is a directory <DIR>
- Actual name of the directory or file



Task 3: To use CD command
-Type **CD desktop** at command prompt **C:\>** (root directory).
-The prompt changes to **C:\Users\smusingo\Desktop>** (Desktop directory).



Task 4: To use CD.. Command (CD with 2 dots)
-Type CD.. at command prompt **C:\Users\smusingo\Desktop>**
-The prompt changes to **C:\Users\smusingo>**
-It moves into the inner user directory **C:\Users\smusingo**
-Type CD.. Again at command prompt **C:\Users\smusingo>**
-The prompt changes to **C:\Users>**
-It moves again into the inner root directory **C:\Users**



Task 5: To use CD\ command (CD with a back slash)
-Type CD\Windows at command prompt **C:\Users\smusingo\Desktop>**
-The prompt changes to **C:\>** (root directory).
Task 6: To change to another drive or directory
-Type D: (drive letter D followed by colon) at command prompt **C:\>**

-The prompt changes to **D:\>** (D drive or directory).
Note:
Some commands require administrator accounts and confirmation by the user (e.g. FORMAT, CHKDSK, DISPART). Why?

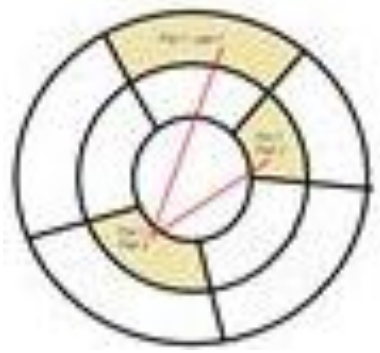
5.4 Formatting HDDs
Formatting refers to preparing a disk for storage. There are 2 types:
1. Low-level formatting
2. High-level formatting

Low-level formatting
-This is **physical formatting** done once in the factory, and involves creating tracks, sectors and cylinders on the platter (disk).
-ECC (Error Correction Codes) are used by the heads to write and read data so that corrupted data can be detected.
High-level formatting
-This is **logical formatting** done by the OS (e.g. DOS, Windows, Linux), and involves creating directories & files on the disk i.e. FAT or NTFS file system created.
-OS initially writes a blank directory on the disk to create order for storing information.
Note:
- High-formatting is part of installation of hard disk drive.
- High-formatting erases/deletes all data stored on the disk.
- MS DOS command for formatting is: **FORMAT** drive**:** (colon) e.g. **FORMAT C:**

Why format a disk?
1. To create directories and files.
2. To store and retrieve data.

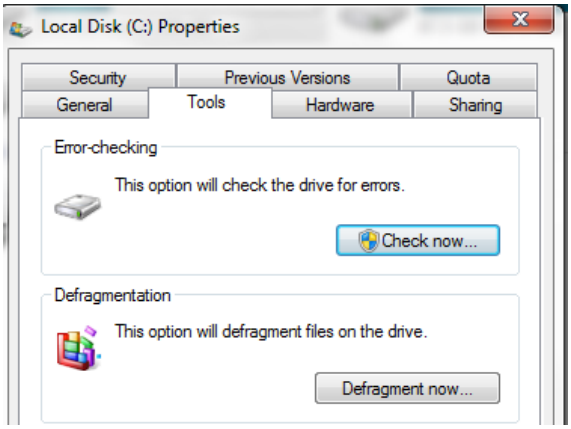
5.5 Optimising HDDs
This refers to making efficient use of the hard disk drive, and includes:
1. Defragmentation
2. Partitioning
3. Disk caching

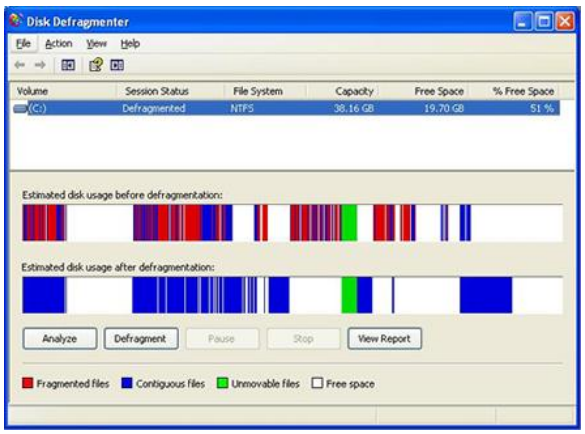
Fragmentation
-This refers to storage of data and programs on the disk in a discontinuous (not adjacent) way.
-File system uses clusters (blocks) of data on the disk, where by each file occupies at least one block.
-Fragmentation occurs on a hard drive when there is no block of free space large enough to contain the complete file.
-With continuous addition and deletion of files, blocks with data will not be contiguous (adjacent) i.e. they will be fragmented (scattered) on the disk.
-Excessive fragmentation wastes disk space and increases access time, since heads have to search for each individual piece of data instead of accessing one large contiguous chunk.
-This affects all files, including system files and application files, which can lead to slow boot times and slow application launches.



Defragmentation

-This involves eliminating fragmentation to improve hard disk drive performance.
-Windows comes with a simple utility called **Windows Defrag** which is used to remove fragmentation.
-Commercial software do exist also and they do the work than Windows Defrag **e.g. Defragmenter Lite/Pro/Plus, Mac OS X, and Drive Genius 3.**
-These should be used on a regular basis to improve performance and save wear and tear on the drive.
Note: NTFS is not prone to fragmentation like FAT.
To defrag hard disk drive
Open **My Computer** >Right click drive > Click **Properties** >Click **Tools** tab.





Partitioning

This involves logically dividing the disk into 2 or more logical drives (C, D, etc.)

Why partition a hard disk drive?

1. To make it easier to organize different kinds of data files.
2. To have multiple operating systems e.g. dual boot (2 operating systems).

Note:

- Some operating systems need to seat on others i.e. installed after others.
  To implement backward compatibility.
- Windows 7 seats on earlier Windows i.e. install Windows 7 after others.
  Linux seats on Windows i.e. install Linux after Windows.

-There are 2 types of partitions:
1. Primary partition
2. Extended partition

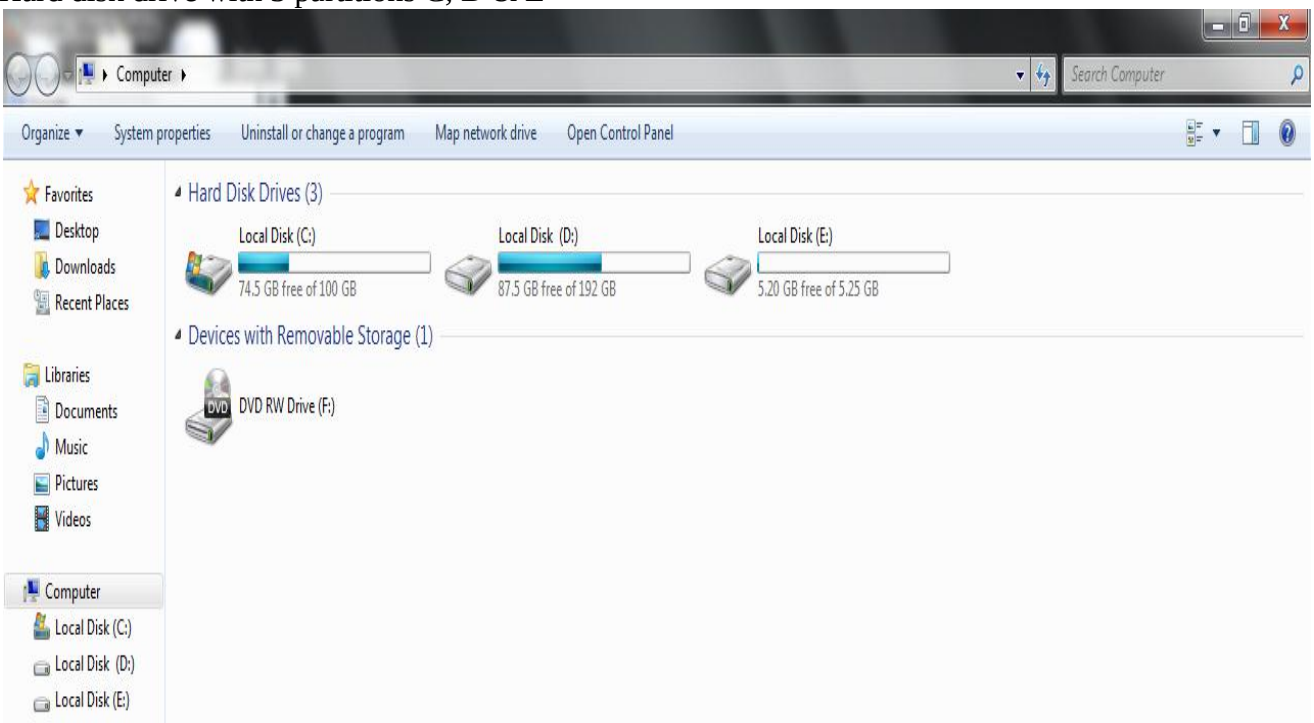
Primary partition

-This is the part of HDD where the **boot information** for the operating system is stored, and it is usually drive C.
-Primary partitions are for storage of the **boot sector**, which tells the computer where to find the operating system.

Extended partition
-This is part of HDD that does not have operating system, and is further divided into logical drives with letters D, E, F, ..., Z (where application).
-Remember, drive letters A and B were for floppy disk drives but have now become obsolete.

Note:
- Newer operating systems can use all of the drive as a single primary partition.
- Partitioning involves formatting which erases/deletes all data stored on the disk.
- MS DOS command for partitioning is: **DISKPART** drive**:** (colon) e.g. **DISKPART C:**

Hard disk drive with 3 partitions C, D & E

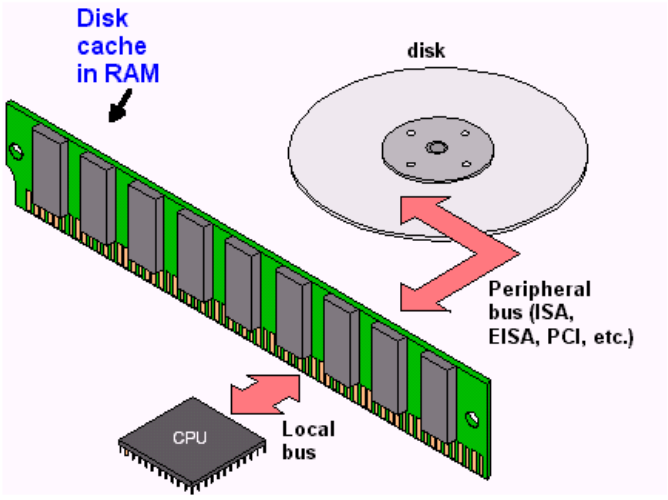


Note: DVD-ROM drive is drive F and partition E.

Advantages of partitioning
1. Reduces cluster size i.e. takes a shorter time to access a file.
2. Each partition is accessed like a different disk
   e.g. You can format a partition without affecting other partitions.

3. Separates data, applications and OS.
   E.g. data in one partition and SW (applications & OS) in another partition.

Disk caching

-Disk caching involves storing copies of frequently used disk sectors in RAM so that they can be read without accessing the slower hard disk.
-This reduces seek time (access time), the time the heads take to read from or write to a hard disk.
-Disk cache may be a specified portion of RAM or buffer memory on a hard drive.
-Disk cache holds data that has recently been read, sometimes adjacent data areas that are likely to be accessed next.

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